



A first calibration of culturable bacterial diversity and their dual resistance to heavy metals and antibiotics along altitudinal zonation of the Teesta River

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Abstract

Culturable bacterial diversity and co-occurrence of heavy metal and antibiotic resistance were investigated from the water and sediments along the course of the Teesta River, in the Eastern Himalayas. Water and sediment samples collected from six sampling points during the monsoon and winter seasons were subjected to analysis of physico-chemical parameters, heavy metal contamination and antibiotic tolerance. The culturable bacterial diversity established by application of bacterial culture and 16S rRNA gene sequencing, ascertained the majority belonged to Proteobacteria, Actinobacteria, Firmicutes, Bacteroidetes and *Deinococcus-Thermus*. Among the 5 phyla, Proteobacteria and Actinobacteria were the dominant phyla present in both water and sediment samples, whereas Bacteroidetes, Firmicutes and *Deinococcus-Thermus* were unique to particular sites. The Shannon index indicated that the bacterial richness was more in the water column as compared to sediment. From the total of 245 isolates, 69 genera were identified. Heavy metal tolerance and antibiotic resistance profiles showed some isolates to be tolerant to high levels of heavy metals and multiple antibiotics indicating a major concern in terms of river ecosystem serving as a pool for dissemination of such resistant genes. The antibiotic resistance and heavy metal contamination diversified along the human-impacted downstream sites, endorsing the contribution of anthropogenic factors. The present report on bacterial diversity and the associated metal and antibiotics tolerance among bacteria is the first of its kind on Teesta River, the only major river system flowing through the state of Sikkim and parts of North Bengal.

Keywords Teesta River · Culturable bacterial diversity · Anthropogenic factors · Heavy metal tolerance · Antibiotic resistance

Introduction

The microbial ecosystems are key components and vital players in aquatic ecosystems, biogeochemical cycling, microbial food webs, degradation of pollutants, and energy flows (Guan et al. 2018). The discharge of wastewater from urban areas and industrial effluents has influenced the bacterial community composition, species abundance and their function, thus disrupting the nutrient cycling processes, negatively affecting the energy flow and propelling loss

of biodiversity. It is an established fact that the structure and function of aquatic ecosystems are facing an existential threat from the changing global environment. Yet, the responses of the microbial community to environmental variations are far from complete, especially since studies of the marine and lake ecosystems have taken precedence over river systems (Zeglin 2015). Spatial and temporal heterogeneity in bacterial diversity has been observed in various rivers driven by several environmental factors like temperature, nutrient uptake, organic matter concentration and composition, hydrology and various land use patterns (Dodds et al. 2000; Mulholland et al. 2008; Boyero et al. 2011). Concentration of metals in the river system was found to be the most prominent driver of microbial ecosystems, whereas nutrients and land use were the least defined. The same study also concluded that temporal variation exhibited a higher percentage of prevalence (93%) than stream (86%) and longitudinal variation (83%) in defining microbiota composition

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and structure in lotic systems. Rivers are contaminated not only with agents like pesticides, heavy metals or pharmaceuticals but also potentially allochthonous microorganisms (Breton-Deval et al. 2019). Therefore, it is also necessary to study the response of microbes at the structure and community level and calibrate the response of microbes to the pollutants.

The Indian rivers are not just economically important, but revered in tradition and culture (Semwal and Akolkar 2006). Escalation in population, industrialization, and unsustainable agricultural practices have rendered rivers vulnerable to pollution from unregulated effluents. Consequently, heavy metal contamination of the riverine ecosystem has been attributed to anthropogenic activities (Singh et al. 2017; Patel et al. 2018; Bhuyan et al. 2019). Elevated levels of toxic heavy metals like Hg, As, Pb, Cd, Cr, Zn and Cu have been reported from various Indian rivers (Mahato et al. 2017; Singh et al. 2017; Patel et al. 2018; Rahman and Singh 2018). Heavy metal-tolerant bacteria isolated from polluted river systems display tolerance to copper, lead, nickel, cobalt, chromium, cadmium and mercury (Rajkumar et al. 2012; Sair and Khan 2018). In response to toxic concentrations of heavy metals, many aquatic organisms, including the microorganisms develop tolerance. This leads to organizational change in bacterial communities, with the resistant and tolerant species becoming more dominant than the native species (Klerks and Weiss 1987).

Studies have revealed that co-selection takes place in environmental bacteria with metal and antibiotics resistance (Wright et al. 2006; Kalayu and Ahemad 2014). Co-selection or co-resistance results from several genes in the same mobile genetic element like a plasmid, encoding for resistance to various antimicrobials. The genetic elements may contain genes for both heavy metals and antibiotics, or the same genetic determinant could confer resistance to both heavy metals and antibiotics known as cross-resistance. Typical example for cross-resistance to various antimicrobials is the multidrug-resistance (MDR) efflux pumps (Hasman and Aarestrup 2002). Upregulation of AcrAB efflux system in *E. coli* due to oxidative stress in the presence of heavy metals like Cr and Cu has been reported to enhance the tolerance of the organism to various antibiotics like oxacillin, novobiocin, tetracycline and chloramphenicol (Harrison et al. 2009). Bacteria in metal contaminated environments have been reported to be a richer source for antibiotics resistance phenotypes, than in control areas (Baker-Austin et al. 2006). In addition, rivers are a major means of dissemination of antibiotic-resistance genes, which is a major public health concern.

The Teesta River system originating in the Eastern Himalayas, once famed for its pristine lotic waters was the focus of our present study. The spatial-temporal variation of the culturable bacterial diversity in surface water and sediment

samples was undertaken along with the co-occurrence of heavy metal and antibiotic resistance. The national water quality monitoring programme has categorized the river based on biological oxygen demand (BOD) values to belong to Priority class 5 with BOD range 3–6 mg/l (CPCB 2018). The Teesta River is flanked by various food, beverage and pharmaceutical companies which could be the possible source of river water contamination in terms of heavy metals, antibiotics and other pharmaceutical products. Therefore, considering the importance of profiling microbial diversity and dynamics in a riverine system which experiences various anthropogenic interventions along its course, the proposed study on the Teesta River could establish a valuable precedence.

Materials and methods

Study area and sampling

River Teesta, a prominent river of the Eastern Himalayan originates in the Teesta Khangse glacier located in north-eastern part of Sikkim in Northeast India. The total catchment area is 12,159 km² and is 414 km long with the greater part in India (151 km in Sikkim, 19 km along the border between Sikkim and West Bengal, 29 km in West Bengal), 94 km along the Indo-Bangladesh border, and the final 121 km in Bangladesh. In terms of socio-economic development, Sikkim has flourished in the past decade with various food, beverage and pharmaceutical industries that have been established along the banks of the river. In addition, Teesta has been used for power generation and has five operational hydro-power projects located in Sikkim and West Bengal. This has led to the destruction of vegetation and natural habitats in these areas. The sampling points and site description are depicted in Fig. 1 and Table 1. Water and sediment samples were collected in pre-sterilized containers during monsoon and winter season in the year 2018–2019 from six sampling sites along the river with the altitude ranging from 5183 to 293 m above mean sea level. Site T1 is a tributary site between site S2 and S3. The samples were brought to the laboratory on ice condition.

Background study of the sampling sites

The physico-chemical parameters and the heavy metal concentration in the river surface water and sediment samples were determined following the standard APHA (1998) protocols. Physico-chemical parameters like temperature, pH, total dissolved solids (TDS), conductivity, dissolved oxygen (DO) and biological oxygen demand (BOD) were analysed. TDS, conductivity and pH were analysed using Eutech Instruments (model CON2700/pH 2700). For DO

Fig. 1 Map of Study area designed using QGIS software version 3.18.2. <https://qgis.org/en/site/forusers/download.html#>

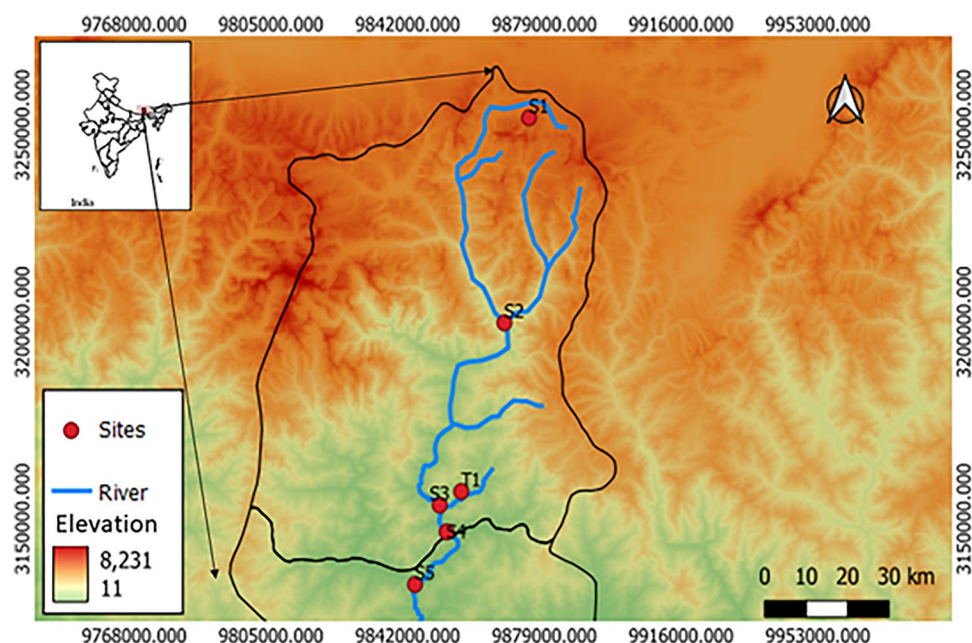


Table 1 Site description, location of the sampling sites and physico-chemical parameters of the water samples

Parameters	S1	S2	T1	S3	S4	S5	BIS limit
Elevation (m)	5183	1790	516	426	333	295	
Coordinates	N28°2.10.2' E088°42.32.6'	N27°36.355' E088°39.062'	N27°15.544' E088°33.118'	N27°13.896' E088°29.517'	N27°10.864' E088°30.027'	N27°04.954' E088°26.654'	–
Site description	Uppermost stretch of the river with minimum human interference	Moderate human settlement with tourism industry and hydro-project in the vicinity	Tributary of river Teesta passing through human settlement and industrial areas	Site with intense human settlement and mining activities	Industrial belt along with human settlement	River banks under forest cover with human settlement and mining activities	–
Temperature	Mn 5 ± 0 Wn 2 ± 0	17 ± 0 9 ± 1.41	18 ± 0 14.5 ± 0.71	18 ± 0 14.5 ± 0.71	18.25 ± 0.53 15 ± 1.41	– 15 ± 01.41	–
pH	Mn 7.4 ± 0.23 Wn 7.26 ± 0.09	7.59 ± 0.07 7.59 ± 0.04	7.71 ± 0.42 7.43 ± 0.35	7.29 ± 0.4 7.79 ± 0.01	7.75 ± 0.07 7.47 ± 0.14	7.53 ± 0.33 7.39 ± 0.15	6.5–8.5
TDS (ppm)	Mn 26.65 ± 0.69 Wn 34.69 ± 0.26	25.2 ± 0.76 35.9 ± 0.67	25.73 ± 5.13 54.04 ± 0.91	28.32 ± 1.3 49.82 ± 2.49	26.62 ± 0.42 44.74 ± 0.78	25.47 ± 0.51 45.18 ± 0.35	500
Conductivity (µS/cm)	Mn 49.92 ± 1.73 Wn 62.64 ± 0.78	47.20 ± 1.59 66.14 ± 0.78	47.11 ± 10.89 99.23 ± 0.9	52.64 ± 2.43 91.72 ± 2.27	49.52 ± 0.93 82.32 ± 0.28	47.45 ± 0.91 85.08 ± 0.48	300
DO (ppm)	Mn 5.32 ± 0.17 Wn 5.05 ± 0.07	6.96 ± 0.79 7 ± 0.28	6.55 ± 0.78 6.38 ± 0.2	7.4 ± 0.06 6.76 ± 0.06	7.36 ± 0.09 6.3 ± 0.42	8.45 ± 0.3 6.38 ± 0.09	5
BOD (ppm)	Mn 0.2 ± 0 Wn 0.29 ± 0.18	0.32 ± 0.11 1.62 ± 0.2	1.05 ± 0.35 2.76 ± 0.45	3.9 ± 0.48 1.96 ± 0.06	0.87 ± 1.32 2 ± 0.57	1.85 ± 1.15 1.18 ± 0.23	5

Mn Monsoon, Wn Winter, BIS Bureau of Indian Standards

estimation, samples were fixed on the site and analysed following Winkler's method. BOD was estimated after 5 days of incubation of the water samples at 20 °C.

Standard protocol of APHA (1998) with slight modifications was employed while preparing the samples for heavy

metal analysis by ICP-OES (Inductively coupled plasma-optical emission spectrometry). 500 ml of water samples from each site was concentrated by heating in a hot plate at optimal temperature to reduce the volume to 50 ml. One gm dry weight of sediment sample was taken in digestion vessel

and 10 ml of metal free distilled water was added, mixed, covered with a watch glass and heated at $95 \pm 5^\circ\text{C}$ for 15 min without boiling. The samples were allowed to cool and 5 ml of concentrated HNO_3 was added and refluxed for 30 min during which brown fumes were given off. The digestion was completed by adding aliquots of 5 ml concentrated HNO_3 until no brown fumes appeared. The sample volume was reduced to 5 ml by heating and cooling after which 2 ml of water and 3 ml of 30% H_2O_2 were added and heated to start the peroxide reaction. 1 ml aliquots of H_2O_2 are added continuously until the effervescence was minimal or the appearance of the sample was unchanged. The sample was heated at $95 \pm 5^\circ\text{C}$ until volume was reduced to 5 ml and after cooling, 10 ml of concentrated HCL was added and refluxed at $95 \pm 5^\circ\text{C}$ for 15 min. The samples were cooled and the particulates of the digestate were removed by filtering through Whatman filter No. 41 and volume was made up to 100 ml with deionized water. All glassware and containers were soaked in 5% nitric acid solution for 24 h and rinsed three times with deionized water before use. The concentration of heavy metals in the samples were analysed by ICP-OES (Thermo Scientific™ iCAP™ 7600).

Bacterial isolation and enumeration

Isolation of bacteria from both the sediment and water samples was done by the standard spread plate technique. In brief, 1 g of sediment sample was suspended in 10 ml of 0.85% saline and kept in shaking condition (150 rpm) for 1 h. The samples were then serially diluted up to 10^{-5} and 0.1 ml from each 10^{-3} , 10^{-4} and 10^{-5} dilutions were spread on Luria–Bertani Agar plates (composition in grams/litre: tryptone- 10, yeast extract- 5, NaCl- 10 and agar- 15). Similarly, water samples were diluted in 0.85% saline up to 10^{-4} fold dilution and subsequently 0.1 ml each from 10^{-2} , 10^{-3} and 10^{-4} were plated on Reasoner's 2A or R2A agar plates (composition in grams/ litre: acicase- 0.50, yeast extract- 0.50, proteose peptone- 0.50, dextrose- 0.50, starch soluble- 0.50, dipotassium hydrogen phosphate- 0.30, magnesium sulphate- 0.024, sodium pyruvate- 0.30 and agar- 15.0). All the media and glassware were sterilized by autoclaving at 121°C for 20 min. The inoculated plates were incubated at 28°C for 24–48 h except for the samples from site S1 from high altitude with temperature range from 2 to 5°C were incubated at 10°C for 2 weeks. After incubation, colonies were selected based on their characteristic colour, morphology and texture, the selected colonies were purified by streak plate technique. The cultures were cryopreserved in the cryovials with 50% glycerol at -20°C . Enumeration of total bacterial load was done by the standard plate count (SPC) technique in the form of colony forming units (CFU). The bacterial load is expressed as CFU/ml in case of water samples and CFU/gm in sediment samples (Kumar et al. 2011).

Bacterial DNA isolation and 16S rDNA amplification

Genomic DNA was extracted using the Bacterial genomic DNA kit by HiMedia, India. The genomic DNA was amplified targeting the 16S rRNA gene using the universal bacterial primers 27F: 5'AGAGTTTGATCCTGGCTCAG 3' and 1492R: 5'TACGGYTACCTTGTACGACT T 3' (Weisburg et al. 1991). The amplification was carried out on a GeneAmp PCR system 9700 (Applied Biosystems, USA). Each 30 μl reaction mixture consisted of 15.75 μl of milli-pore water, 3 μl of 10 $\times$ buffer with 17.5 mM MgCl_2 , 3 μl of each forward and reverse primer of 10 mM concentration, 2 μl of dNTP mixture, 0.25 μl of Taq polymerase and 3 μl of template DNA. The amplified PCR products were electrophoresed on 1.2% agarose gel and purified using QIA Quick Gel Extraction Kit (Qiagen, Hilden, Germany) and sequenced at Macrogen Inc, Seoul, Korea. In this study, a total of 245 isolates from both the monsoon and winter samplings were processed for 16S rDNA amplification.

Sequence alignment and phylogenetic analysis

Before phylogenetic analysis, the sequences obtained were visualised by Chromas software (version 2.6.5) and the contigs were aligned using ClustalW which is in built in Molecular Evolutionary Genetics Analysis software, MEGA 7.0 (Tamura et al. 2011). Identification of the bacterial isolates was carried out by comparing the 16S rRNA gene sequence similarity against all known prokaryotic species in the Eztaxon server (Chun et al. 2007). The nucleotide sequences were submitted to NCBI and the accession numbers were obtained. Raw sequence alignment data are provided in Supplementary S1.

Diversity analysis

To compare the culturable bacterial diversity in both the water and sediment samples, samples were analysed both spatially and temporally to understand the effect of altitude gradient, human intervention and seasonal fluctuations. Diversity analysis was assessed by calculating Shannon–Weiner diversity index (H), Simpson's dominance index (D), evenness index (E) and richness index by Margalef.

Determination of MIC (minimum inhibitory concentration) of the bacterial isolates against heavy metals

MIC of the bacterial isolates was determined against four heavy metals Pb, Zn, Cu and Cd by spot plate technique. Stock solutions (10,000 ppm) of metals were prepared from their respective metal salts of zinc sulphate [$\text{ZnSO}_4 \cdot \text{H}_2\text{O}$], lead nitrate [$\text{Pb}(\text{NO}_3)_2$], copper sulphate [$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$] and

cadmium nitrate [$\text{Cd}(\text{NO}_3)_2$]. The working solutions of heavy metal concentrations were prepared from the stock to achieve the final concentration ranging from 50 to 2000 ppm for Pb, Zn, Cu and 1 to 100 ppm for Cd in the media. The isolates were grown overnight in Luria–Bertani broth and 10 μl of the culture corresponding to OD 0.5 at 600 nm was spotted on the metal containing plates and incubated at 28 °C for 24 h. MIC was determined as the lowest concentration of metal that inhibited growth after 24 h (Aleem et al. 2003).

Antibiotic susceptibility test for the isolates

The antibiotic susceptibility of the bacterial isolates was conducted using agar diffusion technique on Mueller–Hinton Agar (MHA) plates following the guidelines of the Clinical Laboratory standard Institute [CLSI (2013)]. In the present study, total 15 antibiotics were used which are as follows: Meropenem (10 mcg), Imipenem (10 mcg), Erythromycin (10 mcg), Ciprofloxacin (5 mcg), Nalidixic acid (30 mcg), Netillin (30 mcg), Novobiocin (30 mcg), Neomycin (30 mcg), Tigecycline (15 mcg), Lincomycin (15 mcg), Rifampicin (5 mcg), Polymyxin B (300 U), Co-Trimoxazole (25 mcg), Ceftazidime (30 mcg), Trimethoprim (5 mcg) using standard antibiotic discs (HiMedia, India). The plates were observed for clear inhibition zone after incubation for 24 h at 28 °C and the diameter of the zone was measured with a zone scale (Tao et al. 2010). The multiple antibiotic resistance (MAR) index for the test isolates was calculated as a/b where a represents the number of antibiotics the isolates are resistant to and b represents the total number of antibiotics used against the isolate (Krumperman 1983). A MAR index value of >0.2 indicates that the isolate is multiple antibiotic resistant, whereas a MAR index value of ≤ 0.2 indicates a very low or negligible antibiotic resistance.

Statistical analysis

The results were analysed using the PAST software version 4.07b. Two-way ANOVA and Tukey's Post Hoc analysis was performed to see any seasonal and site variation in the bacterial loads. Mann–Whitney test was done to find out for any seasonal influence on the concentration of heavy metals in the samples. Pearson's correlation was employed to understand the relationship between various physico-chemical properties and the culturable microbial taxa in both the water and sediment samples.

Results and discussion

Physico-chemical parameters of river water and sediment samples

The mean values along with the standard deviation of the various physico-chemical parameters are presented in Table 1. The temperature of the river water ranged from 2 °C to 19 °C, and the lowest temperature recorded was at site S1 (2 °C) during winter. The mean values of all the parameters were found to be below the permissible Bureau of Indian Standards [BIS (2012)] limits. The highest monsoon BOD values were at site S3 (3.9 ± 0.48) and lowest at site S1 (0.2 ± 0); whereas the winter highest BOD was recorded at site T1 (2.76 ± 0.45) and lowest at site S1 (0.29 ± 0.18). The increased BOD at these sites could be explained by the nutrient input from domestic wastewaters and open dumping of municipal waste on the roadsides or in streams. This is a common practice in the Indian Himalayan Region where solid waste management is challenging due to various factors (Thakur et al. 2021). There are two existing solid waste dumpsites in Sikkim (CPCB 2019) each located near site T1 and S3. Therefore, unchecked dumping of waste in the streams can certainly accrue detrimental effects on the river ecosystem. The determined DO at S1 is lower than the other sites because of the high altitude with low atmospheric oxygen.

The estimations of Pb, Zn, Cu and Cd were carried out in both the water and sediment samples. The mean concentration of metals for samples analysed is presented in Table 2. The concentration of Pb in the water samples was marginally above the permissible BIS limit at few sites. During the monsoon, Pb concentration was highest at S4 (0.0330 ± 0.021) and lowest at S1 (0.0048 ± 0.004); whereas in winter, the highest concentration was observed at T1 (0.014 ± 0.006) and lowest at S1 (0.0029 ± 0.003). The concentrations of other heavy metals in water were found to be below the threshold range of BIS. In the sediment samples, the concentrations of heavy metals were found to be higher than the Indian River Standard (Subramanian et al. 1987). In the present study, the concentration of Pb and Zn was found to surpass the IRS standards at few sites. The highest concentration of Pb in monsoon was recorded at S4 (27.4 ± 0.017) and lowest at S1 (3.6 ± 0.002), whereas in winter, the highest concentration was recorded at S5 (21.91 ± 0.048) and lowest at S1 (5.6 ± 0.004). Similarly, Zn concentration in monsoon was highest at S5 (85.6 ± 0.026) and lowest at S1 (37.4 ± 0.018), whereas highest concentration in winter was recorded at S4 (6 ± 0.006) and lowest at S1 (1.8 ± 0.008). Investigations into heavy metal contamination of river water have been conducted in many rivers

Table 2 Total viable count (CFU) and heavy metal concentration in the sites during monsoon and winter season

Sampling sites	Seasons	Average CFU/ml in the samples CFU $\times 10^4$	Heavy metal concentration (ppm) in water samples			
			Lead	Zinc	Copper	Cadmium
S1	Mn	3.1 $\pm$ 0.35	0.0048 $\pm$ 0.004	0.0022 $\pm$ 0.004	0.0009 $\pm$ 0.000	0.0002 $\pm$ 0.000
	Wn	2.8 $\pm$ 0.9	0.0029 $\pm$ 0.003	0.002 $\pm$ 0.003	0.0005 $\pm$ 0.003	0.0001 $\pm$ 0.000
S2	Mn	15.56 $\pm$ 2.6	0.0063 $\pm$ 0.003	0.0139 $\pm$ 0.008	0.0014 $\pm$ 0.003	0.0003 $\pm$ 0.000
	Wn	7.6 $\pm$ 0.9	0.0086 $\pm$ 0.003	0.0097 $\pm$ 0.007	0.0015 $\pm$ 0.004	0.0002 $\pm$ 0.000
T1	Mn	28.8 $\pm$ 3.4	0.0161 $\pm$ 0.008	0.1007 $\pm$ 0.016	0.0237 $\pm$ 0.018	0.0009 $\pm$ 0.000
	Wn	11 $\pm$ 1.78	0.014 $\pm$ 0.006	0.0846 $\pm$ 0.009	0.0245 $\pm$ 0.014	0.0005 $\pm$ 0.000
S3	Mn	19 $\pm$ 1.35	0.0082 $\pm$ 0.004	0.0264 $\pm$ 0.005	0.0041 $\pm$ 0.004	0.0004 $\pm$ 0.000
	Wn	7.7 $\pm$ 0.9	0.0033 $\pm$ 0.004	0.0063 $\pm$ 0.002	0.0017 $\pm$ 0.002	0.0002 $\pm$ 0.000
S4	Mn	25.2 $\pm$ 6.36	0.0330 $\pm$ 0.021	0.0414 $\pm$ 0.002	0.0056 $\pm$ 0.002	0.0007 $\pm$ 0.000
	Wn	9.7 $\pm$ 2.12	0.0046 $\pm$ 0.016	0.007 $\pm$ 0.005	0.0024 $\pm$ 0.000	0.0003 $\pm$ 0.000
S5	Mn	24.1 $\pm$ 2.5	0.0149 $\pm$ 0.014	0.0490 $\pm$ 0.0169	0.0102 $\pm$ 0.003	0.0003 $\pm$ 0.000
	Wn	8.5 $\pm$ 0.9	0.0112 $\pm$ 0.009	0.0737 $\pm$ 0.021	0.0172 $\pm$ 0.015	0.0006 $\pm$ 0.000
BIS limit		–	0.01	15	1.5	0.003
Sampling sites	Seasons	Average CFU/gm in the samples CFU $\times 10^4$	Heavy metal concentration (ppm) in sediment samples			
			Lead	Zinc	Copper	Cadmium
S1	Mn	4.6 $\pm$ 0.9	3.6 $\pm$ 0.002	37.4 $\pm$ 0.018	5.4 $\pm$ 0.002	0.04 $\pm$ 0.000
	Wn	4.4 $\pm$ 1.3	5.6 $\pm$ 0.004	1.8 $\pm$ 0.008	1.5 $\pm$ 0.003	0.2 $\pm$ 0.000
S2	Mn	96 $\pm$ 28.6	8.8 $\pm$ 0.007	53.4 $\pm$ 0.014	8.4 $\pm$ 0.002	0.06 $\pm$ 0.000
	Wn	141 $\pm$ 30.3	7.7 $\pm$ 0.003	3 $\pm$ 0.015	2 $\pm$ 0.004	0.6 $\pm$ 0.000
T1	Mn	115 $\pm$ 27.5	11.7 $\pm$ 0.012	55 $\pm$ 0.010	9.4 $\pm$ 0.008	0.17 $\pm$ 0.001
	Wn	156 $\pm$ 11	19.56 $\pm$ 0.042	3.7 $\pm$ 0.003	3.2 $\pm$ 0.002	0.6 $\pm$ 0.000
S3	Mn	123 $\pm$ 20.4	7.2 $\pm$ 0.005	54.4 $\pm$ 0.011	10.8 $\pm$ 0.005	0.10 $\pm$ 0.000
	Wn	163 $\pm$ 23.0	12.5 $\pm$ 0.003	2.6 $\pm$ 0.006	3.4 $\pm$ 0.012	0.5 $\pm$ 0.000
S4	Mn	165 $\pm$ 9.5	27.4 $\pm$ 0.017	83 $\pm$ 0.012	12.7 $\pm$ 0.015	0.65 $\pm$ 0.004
	Wn	175 $\pm$ 22.0	14.8 $\pm$ 0.006	6 $\pm$ 0.017	9.2 $\pm$ 0.006	1.2 $\pm$ 0.001
S5	Mn	111 $\pm$ 25.5	21.0 $\pm$ 0.042	85.6 $\pm$ 0.026	11.2 $\pm$ 0.007	0.21 $\pm$ 0.001
	Wn	155 $\pm$ 8.0	21.91 $\pm$ 0.048	3.7 $\pm$ 0.003	3.2 $\pm$ 0.002	0.6 $\pm$ 0.000
IRS limit		–	11	16	28	–

The values represent the average of three replicates with standard deviation

Mn Monsoon, *Wn* Winter, *IRS* Indian river system

across India. High concentrations of Zn (24.71 mg/l) and Pb (0.13 mg/l), mainly due to anthropogenic contributions have been reported from Kali River in Uttar Pradesh (Mishra et al. 2017). Similarly, Hejabi et al. (2011) reported very high concentration of Zn (54 mg/l) from Kabini river water, in Karnataka. Likewise heavy metal contamination of river bed sediments has been reported from rivers like the Ganga (Singh et al. 2003), Gomti (Singh et al. 2005) Swarnamukhi (Patel et al. 2018) in India.

The mean concentration of Pb in the downstream sites S4, S5, and T1 ranged from 12.5 to 27.4 ppm which surpassed the 11 ppm limit set by BIS. The concentration of Zn ranged from a low of 1.8 ppm in the winter, to very high levels of 85.6 ppm in the monsoon, against the BIS

norm of 16 ppm. These elevated levels of heavy metals in the downstream sites could be attributed to the effects of industrialization. The skewness of data generated necessitated application of Mann–Whitney *U* test, to detect seasonal variations in the heavy metal concentrations in monsoon and winter season. The test revealed no significant differences in the heavy metal concentrations in monsoon and winter season for water samples ($p > 0.05$). In the case of sediment samples, significant variation in the concentration of Zn and Cu was found in terms of season ($p < 0.01$), where lower concentrations of Zn and Cu were detected in winter season at various sites. However, no significant differences were observed in the concentrations of Pb and Cd between the two seasons ($p > 0.05$).

Total viable count

In water samples, the mean bacterial load was highest at site T1 (monsoon- 28.8 ± 3.4 ; winter- 11 ± 1.78) and lowest at S1 (monsoon- 3.1 ± 0.35 ; winter- 2.8 ± 0.9). In the sediment samples, highest bacterial load was found at S4 (monsoon- 165 ± 9.5 ; winter- 175 ± 22.0) and lowest at S1 (monsoon- 4.6 ± 0.9 ; winter- 4.4 ± 1.3). The lower bacterial load at S1 could be explained by the high altitude (5183 m) and cold temperatures (5°C in monsoon and 2°C in winter) in addition to the site being oligotrophic. Out of the water and sediment samples, rich bacterial abundance was observed in the sediment samples in comparison to the water column, as depicted in Table 2. Similar findings of higher bacterial load in the sediment samples than water samples were reported by Niewolak (1998); Rodrigues et al. (2011); Vignesh et al. (2014); further, McLellan et al. (2003), reported enhanced survival of viable bacteria attached to sediments. The high downstream bacterial load was attributed to bacterial agglomeration, increased organic matter deposition in the sediment, and anthropogenic activity (Vignesh et al. 2014; Gardade and Khandeparker 2017). The similar profiles of increasing bacterial load in the downstream sites, and the higher bacterial count during the monsoon from the water samples analysed are supported by studies conducted by Kistemann et al. (2002) and Vignesh et al. (2014). The higher count may be attributed amplification of the microbe population by organic matter, surface run-off water, and faecal matter (from human and animal sources) (Rodrigues et al. 2011). Moreover, the adherence of bacteria to the sediment and their subsequent resuspension in the water are dependent on the river discharge and tides (Gardade and Khandeparker 2017). On the basis of colony morphology, a total of 129 and 116 bacterial isolates were obtained both from water and sediment samples during monsoon and winter season respectively.

Two-way ANOVA with and without interaction was run to look for any bacterial load variation amongst the sites and between the seasons. We found that both the main effect and interaction effect were significant ($p < 0.001$) for both the water and sediment samples. The results showed that the bacterial load was significantly different in the two seasons for both the water and sediment samples ($p < 0.001$). Tukey's post hoc analysis for water samples showed that the bacterial loads were significantly different within sites except sites S2 and S3 ($p = 0.85$); T1 and S4 ($p = 0.56$); T1 and S5 ($p = 0.18$); S3 and S5 ($p = 0.38$); S4 and S5 ($p = 0.97$) which were insignificant. Similarly, Tukey's post hoc analysis for the sediment samples revealed that the bacterial load at site S1 was found to be significantly different from other sites ($p < 0.001$), S2 was significantly different from S4 ($p < 0.01$) and S4 was different from S5 ($p < 0.05$), whereas other sites did not show any significant bacterial load difference.

Culturable bacterial diversity of river water and sediment samples

Based on the colony morphology and 16S rRNA gene sequencing, a total of 129 isolates (water-86 and sediment-43) and 116 isolates (water-61 and sediment-55) were recovered in monsoon season and winter season respectively. The bacterial isolates belonged to 5 major phyla, the Proteobacteria, Actinobacteria and Firmicutes, Bacteroidetes and *Deinococcus-Thermus*. These phyla were represented by various classes belonging to Alpha-proteobacteria, Beta-proteobacteria, Gamma-proteobacteria, Actinobacteria, Bacilli, Flavobacteriia, *Deinococci*, Sphingobacteriia and Cytophagia. Representative comparison of different phyla and classes in both water and sediment samples are shown in Fig. 2. In the water samples, Proteobacteria (51%) and Actinobacteria (31%) were the dominant phyla across sites during both monsoon and winter season. Firmicutes were found to be prevalent during monsoon season than in winter at all the sites. *Deinococcus-Thermus* was a unique phylum that was detected only at site S5, which represented 5% of the total phyla at that particular site. The common classes present in water samples during both the seasons are the Alpha-proteobacteria, Beta-proteobacteria, Gamma-proteobacteria and Bacilli. Flavobacteriia and *Deinococci* were detected only during monsoon season, whereas Sphingobacteriia and Cytophagia were two unique classes found only at site S1, the high altitude site. The Sphingobacteriia and Cytophagia phyla are associated with freshwater habitats and positive correlation with altitude is associated with the abundance of Sphingobacteriia (Siles and Margesin 2016). In the sediment samples, Proteobacteria (48%), Actinobacteria (34%) and Firmicutes (15%) are the phyla commonly present in all the sites during both monsoon and winter seasons. *Deinococcus-Thermus* was present only at site S4 in monsoon season, whereas in winter season, Bacteroidetes was present only at site S1. At the class level, Alpha-proteobacteria, Beta-proteobacteria and Gamma-proteobacteria, Bacilli and Actinobacteria are commonly present at all the sites; Flavobacteriia was detected only at site S1 in both seasons, whereas *Deinococci* was found only in monsoon season and present only at site S4. Similar high abundance of Proteobacteria, Actinobacteria, Firmicutes and Bacteroidetes was reported from the river water and polluted ponds, estuary sediments (Sun et al. 2013; Tai et al. 2020). These phyla are largely distributed in the environment including nutrient enriched waters as they are able to utilise wide range of organic compounds (Horner-Devine et al. 2004). *Deinococcus-Thermus*, generally representing a minor phylum (Janssen 2006) was identified only in the S4 and S5 downstream sites, extremophilic phylum isolated from diverse habitats like arid soils, radioactive sites, nuclear contaminated sediments and water (Theodorakopoulos et al. 2013).

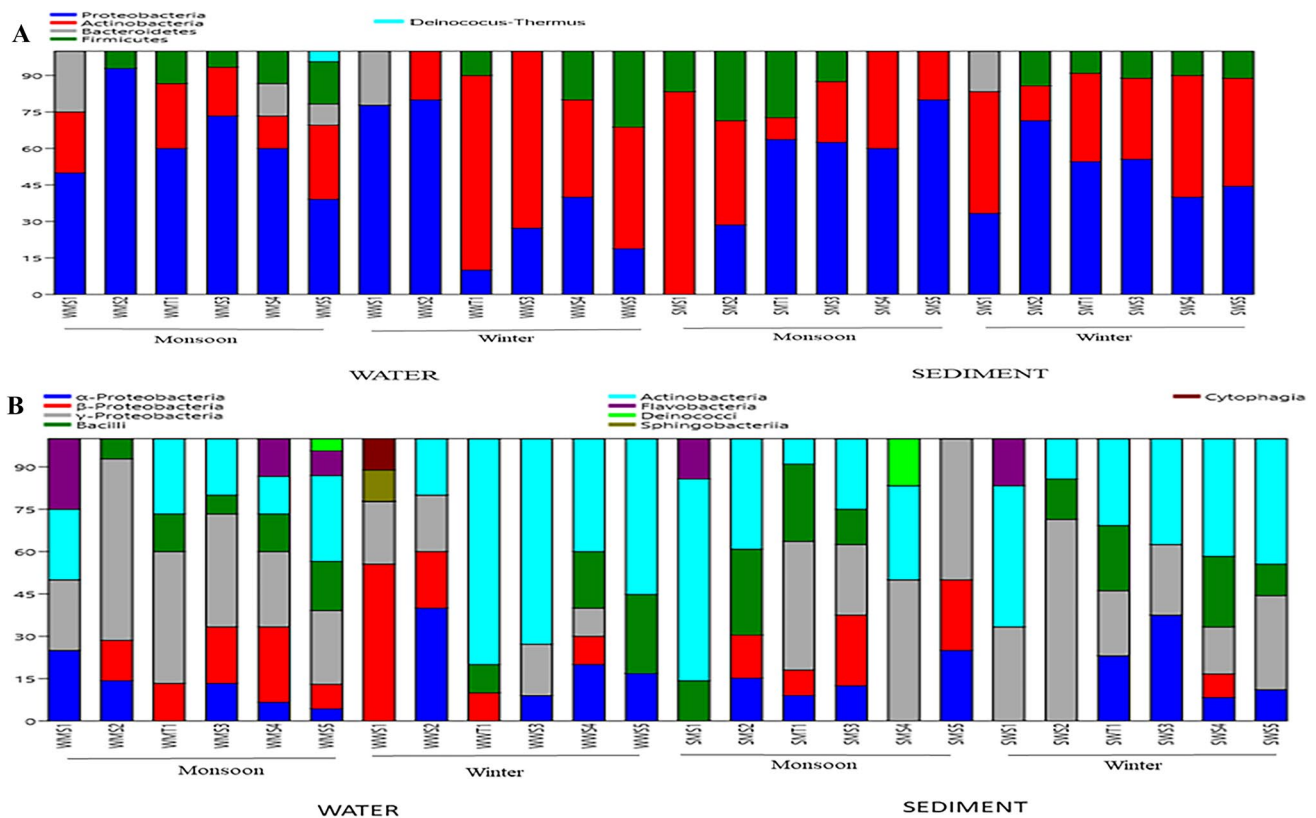


Fig. 2 Graphical representation of bacterial diversity at phylum (A) and class (B) level for the water and sediment samples

Bacterial diversity at the genus level

The total of 245 total isolates collected from all samples, 69 genera were identified after comparing the 16S rRNA sequence with the known sequences deposited in the public database EzBioCloud, yielded <https://www.ezbiocloud.net/identify>. The spatio-temporal comparison of the various genera isolated from water and sediment samples is provided in the Fig. 3. Most isolates exhibited > 97% similarity to the previously deposited sequences. High bacterial diversity was displayed by both the water and sediment samples. The recovered bacterial genera were higher in the water samples as compared to the sediment samples.

In case of the 43% or 138 genera identified in the isolates from water samples, *Pseudomonas* (13%) followed by *Microbacterium* (5.79%), *Bacillus* (5%), *Acinetobacter* (5%), *Brevundimonas* (4.34%), *Serratia* (3.6%) *Arthrobacter* (2.8%), and *Janibacter* (2.8%) were prevalent in more than two sites.

The rest 57.67% isolates represented genera that were uniquely present at less than two sites. Genera like *Flavobacterium*, *Duganella*, *Pedobacter*, *Deefgea*, *Arcicella* were present only at site S1 belonging to the phylum Bacteroidetes and Proteobacteria. Genera distinct to site S2 are *Haemophilus*, *Ideonella*, *Roseateles* and *Variovorax*, whereas

Phycoccus, *Nocardioides*, *Asticcacaulis* and *Vogesella* were unique only to site S3. *Myroides* and *Comamonas* were only present at site S4, and *Priestia*, *Deinococcus*, *Phenylobacterium* and *Brucella* were unique to site S5; while *Dermacoccus*, *Agromyces* and *Gordonia* were unique genera present only at site T1. In the sediment samples, a total of 95 genera were identified amongst which *Pseudomonas* (11.57%) was prevalent in more than two sites followed by *Microbacterium* (10.5%), *Acinetobacter* (8.4%) and *Arthrobacter* (6.3%) accounting for 36.9% of the total isolates in the sediment samples and the remaining 60.1% accounted for the less prevalent genera. *Cryobacterium* was found only in the site S1 which is characterised by high altitude and cold temperature throughout the year. *Commamonas* was identified only from the samples of site S2 and *Deinococcus* was found to be present only at site S4, while, *Hydrogenophaga*, *Paracoccus* and *Ensifer* were unique genera found at site T1. SFig. 1 presents the major genera isolated in water and sediment samples.

The downstream sites impacted by human intervention were represented more frequently by the opportunistic pathogens like *Serratia*, *Acinetobacter*, *Pseudomonas*, *Staphylococcus*, *Microbacterium*, *Gordonia*, and *Brucella*. Occurrence of pathogenic bacteria mainly belonging to Gamma-proteobacteria in the water and sediment associated

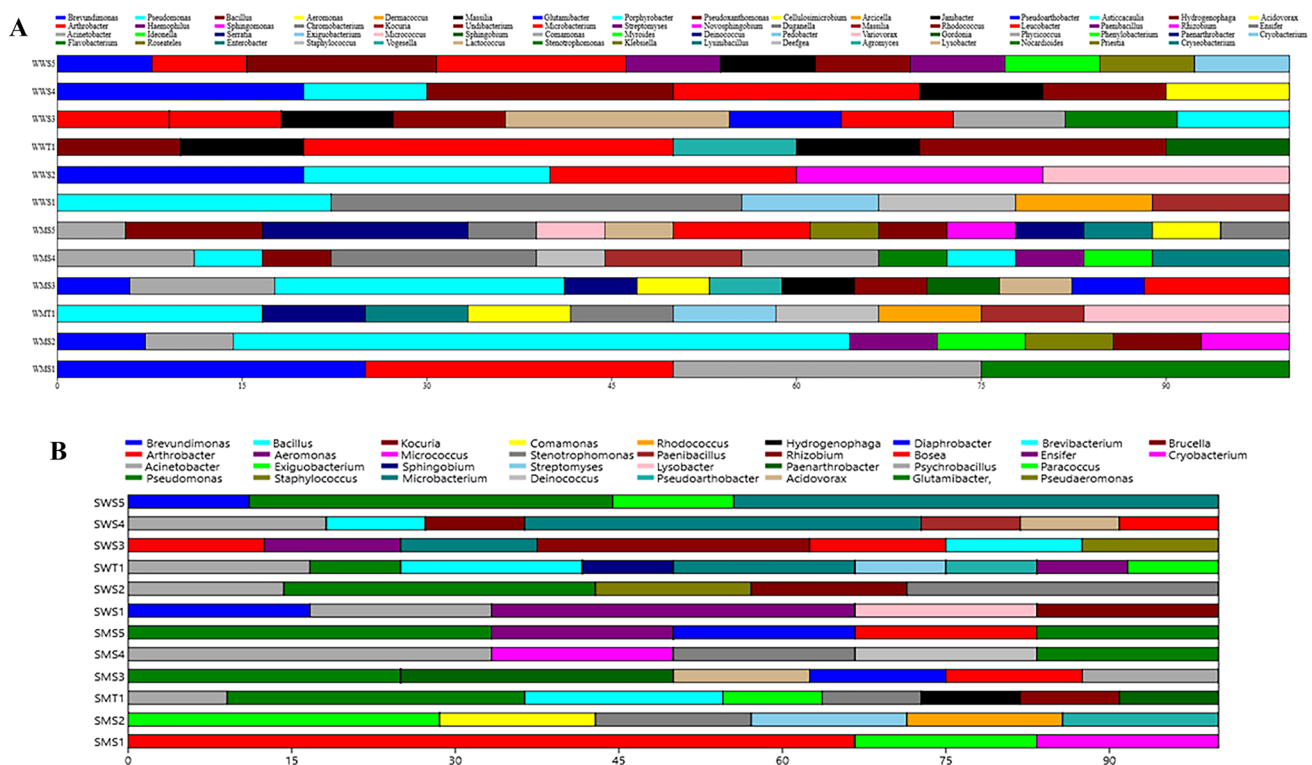


Fig. 3 Graphical representation of various genera identified in water and sediment samples

to higher pollution has been reported (Rodrigues et al. 2011; Gardade and Khandeparker 2017). Water is an easy means of dissemination of these pathogens which come into contact with humans when the water is used for various purposes including recreation.

Diversity analysis

Diversity indices are calculated to compare the community richness and evenness within the various sampling sites and seasons. Shannon–Weiner and Simpson’s index give idea about community composition. Shanno–Weiner index puts more weight on species richness, whereas the Simpson’s index considers species evenness (Kim et al. 2017). The Shannon richness in water samples ranged from 1.76 to 2.91 and 1.96 to 2.74 in monsoon and winter respectively. Similarly in the sediment samples, the Shannon index ranged from 1.03 to 2.29 and 1.38 to 2.47 in monsoon and winter respectively. Shannon index values showed that the bacterial diversity was comparatively higher in the water samples as compared to the sediment samples, which was also apparent from the Shannon rarefaction curve (Fig. 4). In case of water samples, bacterial diversity as indicated by the Shannon index increased with downstream progression, but however no such observation was found in the case of sediment samples. The Simpson’s index ranged from 0.77 to 1 and 0.89 to

1 respectively in monsoon and winter seasons in the water sample. In the sediment samples, the values ranged from 0.6 to 0.95 in monsoon and 0.75 to 0.96 in winter season. The Margalef richness index for water samples ranged from 2.16 to 4.50 in monsoon and 1.37 to 2.89 in winter. According to the Margalef index calculated for the water samples, high bacterial richness was observed at site S5 (downstream site) and least at site S1 (the uppermost site). Similarly, in the case of the sediment samples, the values ranged from 1.12–2.92 in monsoon and 1.37 to 3.22 in winter samples. The upstream and downstream sites displayed least and highest bacterial richness respectively, with the exception of lower Margalef value in winter from site S5. The richness at particular sites in the river could possibly be due to ability of bacteria to grow in particular environmental conditions, which are enriched by inputs from human intervention. In direct contrast, the culturable bacterial diversity is limited to those bacteria which can grow in the controlled conditions and media. Therefore, the relative abundance provided by the culturable bacteria does not provide a clear picture of the microbial community, unlike approaches that do not depend on laboratory culture (Venkatachalam et al. 2015). The summary of various diversity indices in water and sediment samples is given in Table 3. Shannon diversity index was used to study any seasonal variation and the *t* test analysis showed no significant variation in diversity between seasons

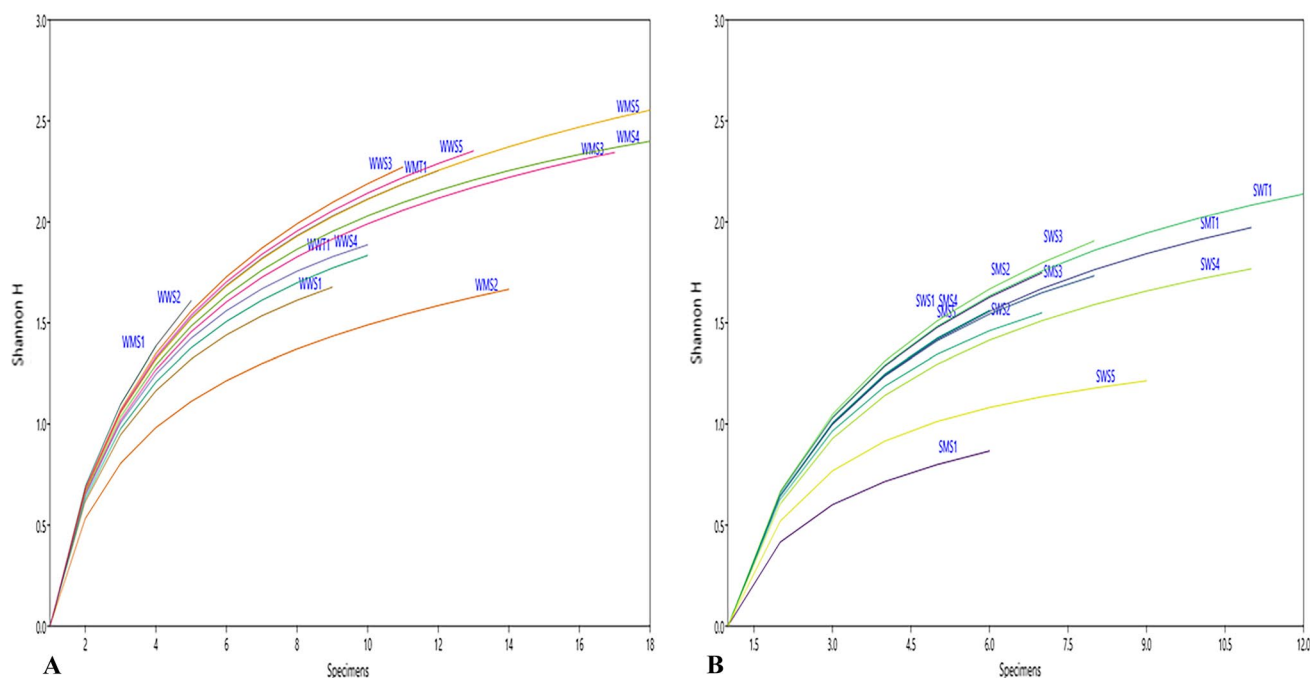


Fig. 4 Rarefaction curves for water and sediment samples

Table 3 Various indices calculated for water and sediment samples

Diversity indices	Water sample											
	Monsoon						Winter					
	S1	S2	T1	S3	S4	S5	S1	S2	T1	S3	S4	S5
Shannon	1.76	1.91	2.63	2.67	2.71	2.91	1.96	2.01	2.13	2.68	2.19	2.74
Simpson	1.00	0.77	0.97	0.94	0.95	0.97	0.89	1.00	0.90	0.98	0.93	0.97
Margalef	2.16	2.65	3.62	3.88	3.81	4.50	2.28	2.50	2.61	3.75	2.61	3.90
Evenness	1.46	0.85	1.39	1.20	1.25	1.32	1.18	1.50	1.21	1.46	1.27	1.40
Diversity indices	Sediment sample											
	Monsoon						Winter					
	S1	S2	T1	S3	S4	S5	S1	S2	T1	S3	S4	S5
Shannon	1.03	2.11	2.29	2.05	1.89	1.89	1.89	1.84	2.47	2.21	2.04	1.38
Simpson	0.60	0.95	0.93	0.93	0.93	0.93	0.93	0.91	0.96	0.96	0.87	0.75
Margalef	1.12	2.57	2.92	2.40	2.23	2.23	2.23	2.06	3.22	2.89	2.50	1.37
Evenness	0.94	1.37	1.24	1.29	1.33	1.33	1.33	1.25	1.32	1.40	1.09	0.99

(monsoon vs winter), in both the water samples ($p=0.547$), and sediments ($p=0.661$) respectively.

Accession numbers for the isolates

Accession numbers for all 245 isolates were obtained after submission of 16S rRNA sequences to the NCBI database <https://www.ncbi.nlm.nih.gov/>. The accession numbers of the isoaltes are as follows: MN726744-MN726749;

MN727110-MN727116; MN733078-MN733085; MN733101-MN733106; MN733214-MN733218; MN733329-MN733339; MN726750- MN726753; MN727117-MN727130; MN733086-MN733122; MN733219-MN733240; MN733340-MN733354; OK090491-OK090506; OK090511-OK090522; OK090551-OK090569; OK090524-OK090545; OK090595-OK090618; OK090572-OK090594. The BLAST search result of the isolates and their accession numbers has been provided in

Supplementary S2 and the number of samples showing identical bacteria has been provided in Supplementary S3.

Correlation analysis

Relationship between species richness and bacterial load with the various physico-chemical parameters was studied by the Spearman's correlation analysis. The species diversity and bacterial load in the water samples showed a positive correlation with each other ($r=0.690$, $p<0.05$). Among all the physico-chemical parameters, only temperature and Cu concentration showed a positive correlation with the species diversity (Table 4); whereas, bacterial load in the water samples showed a positive association with temperature and concentrations of all the heavy metals. Species diversity did not show any significant relationship with the heavy metal concentration in the sediments. However, the bacterial load was positively correlated to Pb ($r=0.664$, $p<0.05$) and Cd ($r=0.813$, $p<0.001$) concentrations (Table 5).

Pearson's correlation analysis was conducted to check for the relationship of the various physico-chemical parameters and bacterial classes. For the water samples, the analysis showed that temperature showed significant ($p<0.05$) positive correlation to the dissolved oxygen and bacterial load, whereas it showed significant negative correlation with the abundance of Sphingobacteria and Cytophagia. Similarly, DO was found to have significant positive correlation with the bacterial load and Deinococci abundance. TDS and conductivity were found exhibit significant variability

with respect to Actinobacteria (positive correlation), and γ -Proteobacteria (negative correlation), as presented in SFig. 2. Heavy metals like Pb and Cd positively correlated with the bacterial load, Zn showed strong positive correlation with Cu and Cd and with the abundance of Bacilli. In one study from metal polluted area, it was found that *Bacillus* sp. was the most dominant and frequently isolated species, which indicates its potential to survive in metal polluted areas and therefore become an indicator of metal contaminated areas (Zampieri et al. 2016).

In the case of sediment samples, Pb and Cd showed significant positive correlation with the bacterial load, and Pb further showed positive correlation with Deinococci. Increased heavy metals have showed higher relative abundance of Deinococci in metal exposed areas. Similarly, Zn and Cu were found to be strongly positively correlated and both metals demonstrated moderate positive correlation with β -Proteobacteria (SFig. 2).

Heavy metal tolerance of the isolates

MICs are crucial in understanding the response of the bacterial isolates to the heavy metals and could be associated with the heavy metal concentration and toxicity (Zampieri et al. 2016). In this study, 107 bacterial isolates were tested for their tolerance to four heavy metals Pb, Zn, Cu and Cd and their MIC values were recorded (STable 1). The bacterial isolates showed varying degrees of tolerance towards the four metals used in this study and the MIC for each metal

Table 4 Spearman's correlation between the Shannon species diversity for water (SDW) and bacterial load for water (BLW) with different physico-chemical parameters and heavy metal concentration in the water samples

	SDW	BLW	Temp	pH	TDS	Pb	Zn	Cu	Cd
SDW									
Correlation coefficient	1.000	0.690*	0.675*	0.294	0.063	0.503	0.517	0.643*	0.504
Sig. (2-tailed)	–	0.013	0.016	0.354	0.846	0.095	0.085	0.024	0.095
BLW									
Correlation coefficient	0.690*	1.00	0.806**	0.357	– 0.175	0.795**	0.771**	0.830**	0.814**
Sig. (2-tailed)	0.013	–	0.002	0.254	0.586	0.002	0.003	0.001	0.001

*Significant at 0.05 level (2-tailed)

**Significant at 0.01 level (2-tailed)

Table 5 Spearman's correlation between the Shannon species diversity for sediment (SDS) and bacterial load for sediment (BLS) with various physico-chemical parameters and heavy metal concentrations

	SDS	BLS	Pb	Zn	Cu	Cd
SDS						
Correlation coefficient	1.000	0.239	0.120	0.021	0.070	– 0.081
Sig. (2-tailed)	–	0.454	0.711	0.948	0.828	0.802
BLS						
Correlation coefficient	0.239	1.000	0.664*	– 0.042	0.217	0.813**
Sig. (2-tailed)	0.454	–	0.018	0.897	0.499	0.001

*Significant at 0.05 level (2-tailed)

**Significant at 0.01 level (2-tailed)

ranged from Pb(500–2000 ppm), Zn (100–1000 ppm), Cu (250–1000 ppm) and Cd (2.5–100 ppm). Most of the the isolates from the downstream sites showed higher MIC values as compared to those isolated from uppermost pristine site (S1). Furthermore, majority of the isolates (82%) displayed higher tolerance towards Pb (MIC value 2000 ppm), and in the case of Zn, 42% showed MIC of 500 ppm and only 13% showed MIC of 1000 ppm; whereas for Cu, 67% displayed MIC of 500 ppm and only 8% showed MIC of 1000 ppm. Determination of MIC for Cd was carried out at lower concentrations (1–100 ppm) as compared to other metals because Cd concentration (0.02–1.2 ppm) was found to be lower in the sites under study. However, most of the isolates (82%) were found to have MIC above 50 ppm. Co-resistance in metal tolerance was observed by Rahman and Singh (2018) where mercury tolerant bacteria from the contaminated river Yamuna water, displayed 100% tolerance to other metals like Pb and Cu. The resistance of bacteria to different metals could be explained by the mechanism of co-resistance where genetically linked factors are expressed simultaneously; or cross-resistance, where the same factor or gene is responsible for resistance to several antimicrobials (Baker-Austin et al. 2006). Therefore, preliminary studies on the occurrence of metal transporting genes were carried out by amplification of the same in some predominant isolates. Current finding indicates that these isolates harbour metal transporting genes belong to the family of RND-HME and P-type ATPase (unpublished data).

Antibiotic resistance profile of the bacterial isolates

All 107 isolates were tested for antimicrobial resistance against 15 different classes of antibiotics, and the isolates showing resistance are summarised in STable 2. The frequency of the antibiotics to which bacterial isolates are resistant, increases from upper to the lower stretch of the

Teesta as depicted in Fig. 5; a clear endorsement of the anthropogenic impact on the river. Considering the multiple antibiotic resistance of an isolate, MAR (multiple antibiotic resistant) index was calculated for each isolate. The MAR index range for each site and the percentage of MAR bacteria at each site are presented in STable 3. The range of the MAR index at each site was 0.13–0.2 at S1, 0–0.46 at S2, 0.2–0.53 at S3, 0.13–0.53 at S4, 0.2–0.46 at S5 and at T1 it ranged from 0.13 to 0.53. An increase in the number of isolates showing multiple antibiotic resistance was observed at the downstream sites (STable 3). The highest percentage of isolates 92% and 95% were observed at sites S4 and S5 respectively. S4 is a site characterised by human settlement and several pharmaceutical companies along the banks of the river, and S5 is a site downstream to S4. The percentage of MAR bacteria increased downstream along the length of the Teesta River (STable 3).

Amongst the various isolates tested for resistance to antibiotics, *Pseudomonas* and *Acinetobacter* showed the highest MAR values that ranged from 0.2 to 0.6 and 0.2 to 0.53 respectively; followed by *Bacillus* for which MAR index ranged from 0 to 0.46, and *Paenarthrobacter* showed 0.4 MAR index. All the *Serratia* and *Microbacterium* isolates showed MAR index of 0.33, and for *Stenotrophomonas* the values ranged from 0.26 to 0.33, while *Brevundimonas* and *Arthrobacter* showed MAR index range from 0.13 to 0.33. *Exiguobacterium* and *Brevibacterium* showed 0.26, *Kocuria* and *Flavobacterium* showed a low of 0.13 MAR index value, and least value of 0.2 was shown by *Cryobacterium* and *Janibacter*. The presence of heavy metals in the environment causes selective pressure leading to increase in bacterial resistance to heavy metals and antibiotics. Metals like Pb, Zn, Cu and Cd have huge applications in industries, and also enter the environment via various vehicular emissions, batteries, mining activities and gasoline spillage and these

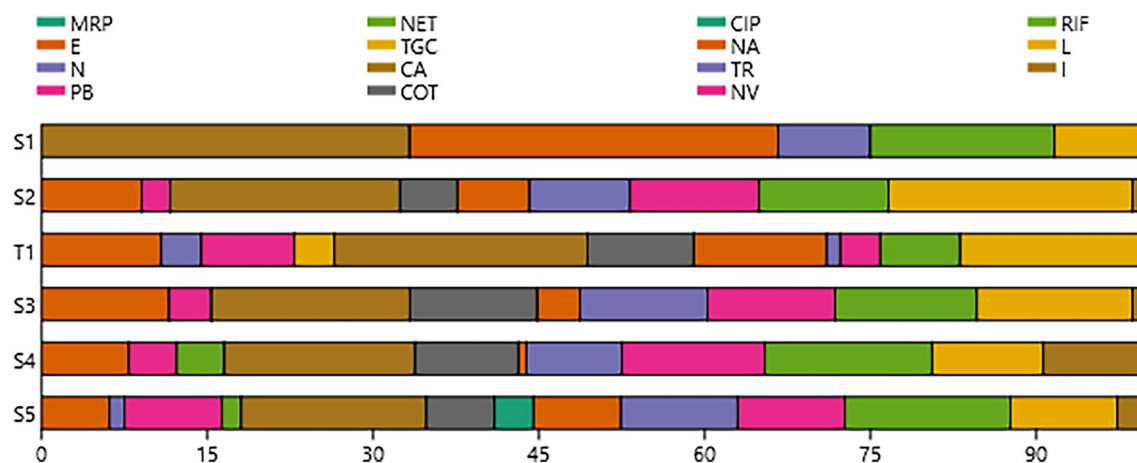


Fig. 5 Graphical representation of the percentage of isolates resistant to various antibiotics at each sites

heavy metals are known to increase the concentrations of selected antibiotic-resistance genes (Mahato et al. 2017). The spread of antibiotic resistance can occur between related and unrelated bacteria, including pathogenic and non-pathogenic ones. Mobile genetic elements, such as plasmids and integrons, encoding the antibiotic-resistance genes are spread among the microorganisms through horizontal gene transfer. Consequently, the genes for resistance are disseminated in the environment and water is an easy means of dissemination (Khan et al. 2013). The MIC values of heavy metals along with inhibition zone for antibiotic disc diffusion assay and the plates are provided as Supplementary S4 and Supplementary S5.

Conclusion

This study showed that the levels of heavy metals like Pb and Zn at certain human-impacted areas tend to be more elevated as compared to the other sites. The bacterial load showed significant variations among the sites with higher load at the downstream sites possibly explained by the anthropogenic organic matter input from adjoining industrial and domestic areas. Bacterial diversity was higher in the water samples though the study found that there was no effect of season on the diversity. Heavy metals alter the bacterial composition and function and here we isolated bacterial strains showing tolerance to both heavy metals and antibiotics and the MAR bacteria increased in the downstream sites. The profile of heavy metal tolerance and antibiotic resistance among the bacteria showed some bacterial isolates to be tolerant to high levels of heavy metals and multiple antibiotics. These findings indicated a major concern in terms of river ecosystem like Teesta serving as a reservoir contributing to dissemination of such resistant genes. The diversification of antibiotic resistance among bacteria and heavy metal contamination along the human-impacted downstream sites of the river endorses the contribution of anthropogenic factors in altering the overall health of the studied river system. The present study which calibrated the culturable bacterial diversity and the associated metal and antibiotics tolerance is the first report of its kind on Teesta River which invites attention of planners and policy-makers to design strategies to protect the once pristine river system from further contamination and deterioration in its quality.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00203-022-02858-1>.

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Author contributions SRJ conceived the study. UC conducted the experiments. SRJ and UC wrote the manuscript and analysed the data.

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Data availability The study did not involve any data deposition.

Declarations

Conflict of interest The authors declare no competing interest.

Ethical approval The study did not require any ethical clearances to conduct the study.

Consent to participate Authors declare their consent to participate in the present work.

Consent for publication Both the authors have consented to publication of the present research findings.

Permission Necessary permission to carry out the study as per the existing regulations was obtained from Department of Forest, environment and Wildlife Management, Office of the Chief Conservator of Forest (T&HQ) cum CWLW, Govt of Sikkim, India.

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